Linguistic Fossils: Recovering Pre-Austronesian Vocabulary Substrates from Nusantaran Languages Using Systematic Subtraction Methodology

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[DRAFT — Paper 8 of VOLCARCH Series — NOT FOR DISTRIBUTION]

> *Series context: Paper 8 of the VOLCARCH series. Paper 7 established the Temporal Overlay Matrix (TOM) — a framework showing that linguistic and genetic clocks indicate earlier human presence in Nusantara than the archaeological record reveals, with the gap correlating with taphonomic loss intensity. Paper 7 answered: when were pre-Austronesian populations present? The present paper asks the complementary question: what did they speak? We apply a vocabulary subtraction methodology — Language_substrate = Modern_language minus Proto-Austronesian_layer — to extract pre-Austronesian lexical residues from modern Nusantaran languages. The Muna language of Southeast Sulawesi, spoken on the island containing the world's oldest securely dated cave art (≥67,800 BP; Oktaviana et al. 2026), is the primary case study. Cross-language substrate intersection across six Sulawesi languages is proposed as a method for partial reconstruction of the language spoken by the world's oldest known artists.*

Abstract

Paper 7 of this series demonstrated that pre-Austronesian populations were present in Nusantara for tens of thousands of years before the Austronesian expansion (~4,000–5,000 BP) — and that the material evidence of their presence has been systematically erased by volcanic burial and coastal submersion. The languages they spoke have been almost entirely replaced by Austronesian languages. Yet language, unlike stone tools or ceramic vessels, leaves a distinctive trace when it is replaced rather than erased: substrate vocabulary. When a speech community adopts a new language, words from the old language frequently survive embedded in the new one, particularly in semantic fields where the incoming language lacked precise terms — local flora and fauna, landscape features, ritual concepts, and body-knowledge accumulated over millennia of local adaptation. This paper operationalizes a vocabulary subtraction methodology for Nusantara: Language_substrate = Modern_vocabulary minus Proto-Austronesian_cognates minus Sanskrit_borrowings minus other_documented_contact_layers. The residual vocabulary — words that cannot be sourced to any known contact language — constitutes candidate substrate: fossilized fragments of the pre-Austronesian languages of Nusantara. We apply this methodology to six Sulawesi languages (Muna, Bugis, Makassar, Tolaki, Wolio, Toraja-Sa'dan) and propose that the intersection of their substrate residuals — words appearing as unexplained residues across multiple languages simultaneously — represents the strongest candidates for genuine pre-Austronesian linguistic fossils. The Muna language receives particular attention: as the language of the island containing the world's oldest securely dated cave art, its substrate residuals are the closest available linguistic approximation to the language spoken by those artists. We further propose four testable predictions about the semantic field distribution of substrate vocabulary that distinguish genuine pre-Austronesian residues from unexplained borrowings or chance resemblances.

Keywords: substrate hypothesis; pre-Austronesian; vocabulary subtraction; linguistic fossils; Muna language; Sulawesi; cave art; language replacement; semantic field analysis; Proto-Austronesian; ABVD

1. Introduction: Language as Taphonomic Medium

The VOLCARCH series has progressively documented how volcanic burial, coastal submersion, historiographic bias, and cosmological displacement have rendered the deep history of Nusantaran civilization invisible. Each of these mechanisms operates on physical or institutional material: sediment buries artifacts, sea water drowns settlements, academic frameworks ignore inconvenient evidence, religious frameworks reframe indigenous knowledge. But there is one medium through which human presence leaves a trace that none of these mechanisms can fully erase: language itself.

When a population is absorbed by or adopts the language of incoming migrants, its original language does not disappear instantaneously. Words — particularly words for locally specific things that the incoming language lacks precise terms for — survive embedded in the new language as substrate vocabulary. These fossilized words are not borrowings in the usual sense: they are not adopted from a prestigious neighboring language by choice, but retained from a displaced ancestral language by necessity. They persist because no adequate replacement exists in the new language, and they cluster in semantic fields that reflect the deepest and most locally specific knowledge of the original population.

Bahasa kuno = bahasa sekarang − bahasa Austronesia

This deceptively simple formula — proposed as an analog to the cosmological subtraction methodology of Paper 5 — operationalizes a well-established concept in historical linguistics: the substrate hypothesis. What is novel in the present paper is its systematic application to the specific problem of recovering pre-Austronesian linguistic content from the languages of Nusantara, combined with a cross-language intersection method for identifying the most robust substrate candidates, and anchored by the extraordinary opportunity presented by the Muna case.

The Muna language is spoken by approximately 300,000 people on Muna Island, Southeast Sulawesi — the island containing Liang Metanduno cave, site of the world's oldest securely dated cave art at ≥67,800 BP (Oktaviana et al. 2026). Modern Muna is an Austronesian language, descended from the same Proto-Austronesian ancestor as Javanese, Malay, and Malagasy. Its speakers arrived on the island approximately 3,000–4,000 years ago — more than 63,000 years after the people who left their handprints on the cave wall. But in that 63,000-year gap, something happened that is the subject of this paper: the incoming Austronesian speakers encountered descendants of those original artists, and some of their words survived the encounter, fossilized in what is now spoken as Muna.

Finding those words — even a fragment of them — would be the closest we can come to hearing the voice of the world's oldest known artists.

2. Theoretical Framework: The Substrate Hypothesis

2.1 Substrate in historical linguistics: established cases

The substrate hypothesis is not speculative — it is a well-established analytical tool in historical linguistics with numerous confirmed applications. When Proto-Indo-European speakers spread across Europe, they encountered existing populations speaking languages that have since been entirely replaced. These languages are now known only through their substrate traces in the daughter Indo-European languages.

The clearest case is the pre-Greek 'Pelasgian' substrate, visible in thousands of Greek place names with characteristic suffixes (-nthos, -ssos, -ssa) that have no Indo-European etymology: Korinthos, Parnassos, Knossos, Hymettos. These are not borrowings from a known neighboring language — they are fossils of the language spoken in Greece before Indo-European arrived. Similarly, pre-Celtic substrate vocabulary is reconstructible from Welsh and Irish words without Indo-European cognates, clustered in semantic fields of local flora, fauna, and topography. Pre-Dravidian substrate is visible in Sanskrit retroflex consonants (sounds absent from all related Indo-European languages but present in Dravidian) — proving that even phonology, not just vocabulary, can carry substrate traces.

The pattern is consistent across cases: substrate vocabulary clusters in semantic fields where the incoming language lacked precise local terms (endemic species, local landscape features, indigenous ritual concepts, body techniques specific to local ecology) and in phonological features that the incoming language's speakers had to adapt to articulate unfamiliar sounds. Both vocabulary and phonology are diagnostic for the present paper's purposes.

2.2 The subtraction methodology formalized

The vocabulary subtraction methodology for Nusantara operates through sequential elimination of known contact layers:

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Language_substrate(L) = Vocabulary(L)
                      - PAn_cognates(L)         [Austronesian layer]
                      - Sanskrit_borrowings(L)  [Indic layer, Paper 6]
                      - Arabic_borrowings(L)    [Islamic layer]
                      - Dutch_borrowings(L)     [Colonial layer]
                      - Malay_borrowings(L)     [Lingua franca layer]
                      - other_documented_contact(L)
                      ──────────────────────────────────
                      = RESIDUAL (candidate substrate)
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The residual set contains three types of items: (1) genuine pre-Austronesian substrate vocabulary — the target of this analysis; (2) undetected borrowings from documented contact languages — a source of false positives that must be controlled; and (3) Austronesian innovations not yet reconstructed in PAn — words that are genuinely Austronesian but whose PAn ancestor has not yet been identified due to gaps in the ABVD reconstruction. Type 3 false positives decrease as PAn reconstruction improves; Type 2 false positives are controlled by the cross-language intersection method described in Section 4.

2.3 Why the cross-language intersection reduces false positives

The key methodological insight of this paper is that genuine pre-Austronesian substrate residuals should appear across multiple Sulawesi languages simultaneously — because all Sulawesi languages displaced the same pre-Austronesian population(s) and therefore inherited substrate from the same source. Undetected borrowings from documented contact languages, by contrast, will have entered different Sulawesi languages at different times through different contact channels and will not systematically co-occur across all languages.

Formally: if word W appears in the substrate residual of languages L1, L2, L3, L4, L5, and L6 — if it cannot be sourced to PAn or any documented contact layer in any of these languages — the probability that it is a genuine substrate fossil increases with the number of languages in which it appears. A word appearing in only one language's residual might be an

undetected borrowing or an internal innovation. A word appearing in the residual of all six Sulawesi languages in our corpus is very unlikely to be an accidental coincidence — it is almost certainly a fossil from a common pre-Austronesian ancestor language of the region.

3. Corpus: Six Sulawesi Languages

3.1 Language selection rationale

Six Sulawesi languages are selected to maximize geographic and phylogenetic coverage of the island while remaining within the computational feasibility of the initial study. The selection criterion is: languages from distinct branches of the Sulawesi Austronesian sub-family, distributed across the major geographic regions of the island, with existing documentation sufficient for Swadesh list and extended vocabulary analysis.

Table 1: Corpus of six Sulawesi languages

Language	Speakers	Region	AUS branch	Key feature	Significance
Muna	~300,000	SE Sulawesi	Muna-Buton	Island isolate	PRIMARY: island of oldest cave art
Bugis	~5,000,000	S Sulawesi	South Sulawesi	Maritime elite	La Galigo corpus; largest language
Makassar	~2,000,000	SW Sulawesi	South Sulawesi	Trade language	Historical trade network hub
Tolaki	~300,000	SE Sulawesi	Celebic	Interior isolate	Geographically closest to Muna
Wolio	~80,000	Buton Island	Muna-Buton	Literary tradition	Same branch as Muna; written tradition
Toraja-Sa'dan	~500,000	C Sulawesi	South Sulawesi	Ritual isolation	High indigenous ritual vocabulary retention

3.2 Data sources

Vocabulary data for each language will be compiled from: the Austronesian Basic Vocabulary Database (ABVD) for Swadesh list items; published dictionaries and wordlists (Muna: van den Berg 1996; Bugis: Matthes 1874; Makassar: Matthes 1859; Tolaki: Mead 1999; Wolio: Anceaux 1987; Toraja-Sa'dan: Friedericy 1933 and more recent documentation); and fieldwork data where available. Sanskrit borrowings will be identified using Zoetmulder (1982) and Monier-Williams (1899). Arabic borrowings via standard Islamic vocabulary lists. Dutch colonial borrowings via Sneddon (2003).

The extended vocabulary beyond the Swadesh-200 list is critical for this analysis. Substrate vocabulary is by definition not in the core vocabulary — if it were, it would have been replaced already, as core vocabulary is most susceptible to replacement pressure. The most productive semantic fields for substrate recovery are the extended vocabulary domains of: (a) local flora and fauna not present in Taiwan or the Philippines (the Austronesian homeland and expansion route); (b) landscape and geological features specific to Sulawesi's volcanic and karst geography; (c) ritual and cosmological terms; (d) fishing and maritime techniques

specific to Sulawesian waters; (e) body-knowledge terms (disease names, medicinal plants, body part terminology beyond the Swadesh core).

4. The Subtraction Pipeline: Methodology

4.1 Step 1: PAn cognate identification

For each word in each language's vocabulary, the first step is identifying whether a Proto-Austronesian cognate exists in the ABVD reconstruction. This is not a simple dictionary lookup — regular sound correspondences must be applied. For example, Proto-Austronesian *mata ('eye') appears in Muna as mata — a straightforward correspondence. But PAn *qaCay ('liver') appears in Bugis as ati after regular sound changes (PAn *q > zero in Bugis; PAn *C > t; PAn *y > i) — a cognate that would not be identified by simple string matching.

The cognate identification pipeline therefore requires: a phonological correspondence table for each language (mapping PAn phonemes to their expected reflexes); a rule-based cognate detection algorithm applying these correspondences; and expert review of the output, distinguishing genuine cognates from chance resemblances (false cognates). The ABVD already provides correspondence tables for most Sulawesi languages; these will be formalized and implemented as a finite-state transducer for computational application.

4.2 Step 2: Contact layer removal

After PAn cognates are identified and set aside, the remaining vocabulary is screened against documented contact layers in sequence: Sanskrit (using Zoetmulder 1982 for Old Javanese, extended to Sulawesi languages via known Indic loanword phonology); Arabic (core Islamic vocabulary, identifying standard Arabic loanwords in Muslim-majority languages including Bugis and Makassar); Dutch (colonial-era borrowings, largely in administrative, technological, and commercial vocabulary); and Malay (widespread Austronesian trade language, borrowings identified by characteristic Malay phonological features absent from the native Sulawesi reflexes of PAn).

Each layer is removed sequentially, with the order chosen to minimize cascade errors: PAn first (largest and most systematic layer), then Sanskrit (next largest in historically Indianized areas), then Arabic and Dutch (smaller and more easily identified by phonological markers), then Malay. The residual after all layers are removed is the candidate substrate set.

4.3 Step 3: Cross-language intersection

The candidate substrate sets for all six languages are compared, and items are classified by the number of languages in which they appear:

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Tier 1 (strongest): appears in substrate of ALL 6 languages
Tier 2 (strong):    appears in substrate of 5 languages
Tier 3 (moderate):  appears in substrate of 4 languages
Tier 4 (weak):      appears in substrate of 2-3 languages
Tier 5 (suspect):   appears in substrate of only 1 language
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Tier 1 and Tier 2 items are the primary analytical focus. These are words for which no PAn ancestor can be identified, no contact language source can be found, and which appear as unexplained residuals in the vocabularies of languages from different branches of the

Sulawesi Austronesian family distributed across the island. The probability that these words are genuine pre-Austronesian fossils is substantially higher than for Tier 4–5 items.

4.4 Step 4: Semantic field profiling

Tier 1–2 substrate candidates are classified by semantic field using a custom tagset developed for this project. The expected distribution of genuine substrate vocabulary — derived from the theoretical predictions in Section 5 — will be compared against the observed distribution as a validity check. If the observed semantic field profile matches the theoretical predictions, confidence in the substrate interpretation is strengthened.

5. Four Testable Predictions

The substrate hypothesis for Nusantara generates four specific, falsifiable predictions about the properties of the substrate vocabulary that should be recoverable by the subtraction methodology. These predictions are derived from the general substrate hypothesis plus specific knowledge of the ecological, cultural, and linguistic context of pre-Austronesian Sulawesi.

5.1 Prediction 1: Semantic field clustering

> *H1: Substrate vocabulary will be significantly non-uniformly distributed across semantic fields, clustering in: (a) endemic flora and fauna; (b) geological and landscape features; (c) ritual and cosmological vocabulary; (d) maritime and fishing technology specific to Sulawesian waters. This distribution will differ significantly from the semantic field distribution of PAn-derived vocabulary, which should be more uniformly spread across all domains.*

This prediction follows from the general substrate theory: incoming languages lack terms for locally specific things, so substrate words survive most in domains of high local specificity. Sulawesi has an extraordinarily high level of biological endemism (80%+ of its mammals are endemic), complex karst and volcanic geology with no parallel in Taiwan or the Philippines, and a maritime ecology specific to the Banda and Flores Seas. All of these are domains where pre-Austronesian populations would have developed precise vocabulary that Austronesian migrants would have needed to borrow.

5.2 Prediction 2: Phonological distinctiveness

> *H2: Substrate vocabulary will show significantly higher rates of phonological features absent from or rare in PAn — particularly retroflex consonants, unusual vowel combinations, and syllable structures incompatible with PAn phonotactics — compared to PAn-derived vocabulary in the same languages.*

When a language community adopts a new language, they carry over phonological habits from their first language. These habits gradually disappear over generations, but words borrowed early in the contact period retain phonological features of the original language. Substrate words are by definition borrowed at the earliest stage of contact — they are the words that survived the initial language shift — and should therefore show the strongest traces of the original phonological system.

This prediction is particularly testable because it makes a phonological claim that is independent of the semantic field claim of H1. Confirming both H1 and H2 simultaneously would be strong evidence for a genuine substrate, as the probability of both patterns occurring by chance is very low.

5.3 Prediction 3: Muna substrate richest among six languages

> *H3: The Muna language will show the highest substrate residual proportion among the six Sulawesi languages — both in absolute count and as a proportion of total vocabulary — reflecting its geographic isolation as an island language with less intense Austronesian contact pressure than mainland Sulawesi languages.*

Island languages are generally more conservative than mainland languages: they are subject to less contact with neighboring languages, experience less prestige pressure from dominant trade languages, and maintain more internal continuity of vocabulary transmission. Muna Island's isolation in Southeast Sulawesi, combined with the relatively late and less intensive Austronesian contact in this region compared to South Sulawesi (the Bugis-Makassar heartland), predicts a richer substrate residual.

The archaeological significance of this prediction is clear: if Muna shows the richest substrate, then the Muna substrate is the richest available source for vocabulary that may preserve traces of the language spoken on Muna Island 67,800 years ago — or at minimum, of the language spoken by the ancestors of the cave artists' descendants before Austronesian replacement.

5.4 Prediction 4: Substrate-cosmology correlation

> *H4: Substrate vocabulary in the ritual and cosmological semantic field will overlap significantly with the 'COSMOLOGICAL-INDIGENOUS' vocabulary identified in Paper 6's feature F6 — and with the ritual residues identified in Paper 5's cosmological stratigraphy analysis. Indigenous cosmological vocabulary and pre-Austronesian substrate vocabulary should be drawing from the same underlying linguistic layer.*

This prediction connects Papers 5, 6, and 8 into a unified framework. Paper 5 identified ritual residues (slametan, danyang, Pranata Mangsa) as surviving fragments of pre-Indic indigenous cosmology. Paper 6 operationalized indigenous ritual vocabulary as feature F6 in the textual phylogenetics framework. The present paper proposes that the vocabulary underlying these residues — the actual words used in pre-Indic indigenous cosmology — should appear as substrate items in the subtraction analysis. If this cross-paper prediction is confirmed, it constitutes a cross-methodological validation: three different analytical approaches (ritual analysis, textual feature extraction, vocabulary subtraction) converging on the same underlying linguistic layer.

6. The Muna Case: Detailed Analysis

6.1 Muna language: background

Muna (also known as Muna-Buton or Wuna by its speakers) is a member of the Muna-Buton subgroup of the Sulawesi Austronesian languages. It is genetically related to Wolio (spoken on Buton Island to the south) and more distantly to the South Sulawesi languages (Bugis, Makassar, Toraja). It has been spoken on Muna Island — an island of approximately 2,963 km² separated from Southeast Sulawesi by a narrow strait — for approximately 3,000–4,000 years based on Austronesian phylolinguistic estimates.

The primary linguistic documentation of Muna is van den Berg (1996), which provides a grammar and dictionary sufficient for the initial phases of this analysis. The language is currently considered stable, with active intergenerational transmission, though it is in contact with Indonesian (the national language) and Wolio. Its isolation as an island language has preserved conservative features absent from closely related mainland languages.

6.2 Known unexplained vocabulary in Muna

Even without a systematic subtraction analysis, Muna linguistic documentation reveals vocabulary items that van den Berg (1996) notes as having no clear Austronesian etymology. These fall into three categories that are consistent with substrate predictions:

Geological and landscape terms: several Muna words for specific karst topography features — types of sinkholes, cave formations, underground water channels — lack PAn cognates. Muna Island is predominantly karst limestone, and its residents have developed precise vocabulary for karst features that Austronesian languages from volcanic origins (Taiwan, Philippines) simply did not possess. These terms are likely substrate survivors preserved because no Austronesian equivalent existed.

Marine life: several Muna fish and marine invertebrate names lack PAn cognates and do not match any known contact language source. The waters around Muna Island are part of the Banda Sea biodiversity hotspot, with endemic species not found in the Austronesian expansion route. Pre-Austronesian populations would have developed precise names for these species over tens of thousands of years; some of those names apparently survived the language shift.

Ritual and ceremonial terms: certain Muna ritual terms — particularly in the context of ancestral spirit communication and cave-related ritual — lack PAn etymology. Given that the caves of Muna Island have been sacred sites for at least 67,800 years (as evidenced by the hand stencils), the possibility that some ritual vocabulary associated with cave use survives from pre-Austronesian times is not implausible.

6.3 The cave art vocabulary question

The most speculative but most evocative question this paper raises is whether any Muna vocabulary is directly connected to the cave art tradition. We do not claim that specific Muna words for 'hand stencil' or 'red ochre' can be identified as pre-Austronesian — the evidence does not currently support such a specific claim. What can be proposed is the following:

If the hand stencil makers of 67,800 BP had words for the caves they painted in, for the pigments they used, for the ritual or communicative purpose of the images, and for the animals depicted — and if their descendants continued to live on Muna Island and use these terms through the period of Austronesian contact — then some fraction of those words may survive in the Muna substrate residual. Not as identifiable 'words for cave art' but as unexplained vocabulary items in semantic fields associated with caves, ochre (iron oxide minerals), and the fauna depicted (anoa, babirusa, and other Sulawesian endemics).

The substrate analysis cannot prove this connection. But it can identify vocabulary items that meet the criteria — pre-PAn, phonologically distinctive, in the relevant semantic fields — and flag them as hypotheses for future investigation through aDNA analysis of cave sediments, archaeological investigation of cave-associated deposits, and comparative ethnohistorical research.

7. Broader Implications: A Nusantara-Wide Substrate Map

7.1 Extension to Java

The subtraction methodology applied in this paper to Sulawesi can in principle be applied to Javanese — and the result would be directly relevant to the VOLCARCH series' primary focus on Java. The formula:

> *Javanese_substrate = Modern Javanese − PAn_cognates − Sanskrit_borrowings − Arabic_borrowings − Dutch_borrowings − Malay_borrowings*

would yield vocabulary that cannot be explained by any documented contact language — candidate fossils from the languages spoken in Java before Austronesian expansion arrived approximately 3,000–4,000 years ago. These would be the words of the people whose material culture is now buried under meters of volcanic sediment (Paper 1) and whose coastal settlements are now under the Java Sea (Paper 2).

The Javanese substrate analysis faces an additional challenge not present in the Sulawesi case: the Sanskrit layer in Javanese is exceptionally thick, and the boundary between Sanskritic Old Javanese Kawi vocabulary and indigenous substrate is harder to establish than in Sulawesi languages with less Sanskrit influence. Paper 6's diglossia framework and the KawiKupas tool proposed there are prerequisites for a reliable Javanese substrate analysis.

7.2 Toward a substrate geographic map of Nusantara

Applying the subtraction methodology systematically across all major Nusantaran language groups — Javanese, Sundanese, Madurese, Balinese, Batak, Minangkabau, the Kalimantan languages, the Sulawesi languages, the Maluku languages, and the Nusa Tenggara languages — would produce a substrate geographic map of Nusantara: a spatial distribution of pre-Austronesian linguistic fossils across the archipelago.

This map would have direct implications for the Temporal Overlay Matrix of Paper 7. If substrate density correlates with genetic 'ghost population' density (regions where modern populations carry more pre-Austronesian genetic ancestry), and if both correlate with the taphonomic loss index, then all three independent lines of evidence — linguistic substrate, population genetics, and taphonomy — would be telling the same story: that the pre-Austronesian populations of Nusantara were denser, more sophisticated, and more widely distributed than the current archaeological record suggests.

7.3 Comparison with confirmed substrate cases worldwide

The credibility of the Nusantaran substrate hypothesis is strengthened by comparison with confirmed cases globally. The methodology proposed here — systematic vocabulary subtraction followed by cross-language intersection — is analogous to the methods used to reconstruct pre-Indo-European substrates in Europe, pre-Dravidian substrates in India, and pre-Bantu substrates in southern Africa. In each case, the substrate hypothesis began as speculation and was confirmed through the accumulation of vocabulary items with consistent phonological and semantic field properties.

The Sulawesi case has one advantage over the European and Indian cases: the pre-Austronesian populations of Sulawesi are better documented genetically than the pre-Indo-European populations of Europe. The genetic signal of Australo-Melanesian ancestry in modern Sulawesian populations provides an independent line of evidence that pre-Austronesian people were indeed present, that they were absorbed rather than entirely replaced, and that their descendants contributed to modern populations. This genetic confirmation means that the substrate hypothesis is not just a linguistic inference but a claim about a population whose existence is independently documented.

8. Conclusions

This paper has proposed a vocabulary subtraction methodology for recovering pre-Austronesian linguistic fossils from modern Nusantaran languages, formalized the cross-language intersection approach for identifying the most robust substrate candidates, and developed four testable predictions about the properties of genuine substrate vocabulary. The Muna language of Southeast Sulawesi — the language of the island containing the world's oldest securely dated cave art — is identified as the primary case study.

The central argument can be stated simply: language is a taphonomic medium. It buries the past differently from sediment and water — not by covering it up, but by absorbing it, metabolizing it, and preserving fragments of it embedded in new structures. The pre-Austronesian languages of Nusantara are gone in the sense that no community speaks them as a primary language. But they are not entirely gone in the sense that their words — some of them — still exist in the mouths of people who do not know they are speaking ancient fossils.

> ***The formula is simple: Bahasa kuno = bahasa sekarang − bahasa Austronesia. What remains is not silence — it is the oldest human voice in Nusantara, waiting to be heard.***

Finding that voice — even a few words of it — would not merely be a contribution to historical linguistics. It would be a contribution to human self-knowledge: evidence that the people who first looked at the walls of a limestone cave 67,800 years ago and decided to press their hands against the rock and blow red pigment around them were people with language, with culture, with names for the world they inhabited. Names that, against all odds, may still be among us.

Draft Notes & Placeholders

[ ] Section 3.2 — van den Berg (1996) is the key Muna source. Verify full citation: van den Berg, R., 1996. A Grammar of the Muna Language. KITLV Press, Leiden. Check if there is a more recent dictionary or wordlist.

[ ] Section 6.2 — The 'known unexplained vocabulary in Muna' section is currently theoretical. Actual examples need to be identified by going through van den Berg (1996) systematically. This is the core empirical work of the paper.

[ ] Section 4.1 — The finite-state transducer for PAn correspondence rules: check if one already exists for Sulawesi languages in the ABVD infrastructure. Contact Simon Greenhill (ANU) who maintains ABVD.

[ ] Section 5.3 — H3 (Muna richest substrate) is the most testable prediction. It could be tested with existing data before the full pipeline is built. Consider whether a 'pilot study' note in this section would strengthen the paper.

[ ] Section 7.1 — Javanese substrate analysis: the Sanskrit layer thickness issue is real. Confirm that KawiKupas (Paper 6) is explicitly framed as a prerequisite here.

[ ] Journal target: This is the most emotionally resonant paper in the series — 'recovering the voice of the world's oldest artists.' Consider Nature Human Behaviour or PNAS for reach. The Muna-cave art connection is the hook for a press release.

[ ] Ethical note: Any research involving Muna community knowledge requires free, prior, and informed consent from the Muna speech community. This should be addressed explicitly in the methods — possibly as a dedicated ethics subsection.

[ ] Consider adding an appendix: 'Candidate substrate items identified in existing Muna documentation' — even a short list of 10-20 items would make the paper more concrete and reviewable.

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[Full literature compilation pending — Sulawesi linguistics, substrate hypothesis methodology, pre-Austronesian genetics literature]